

Personalized Weather Station – Updated Project Definition

High Level Description

The Personalized Weather Station is a compact embedded environmental monitoring system designed to measure and display real-time weather conditions such as temperature, humidity, and atmospheric pressure. The project combines custom PCB development, sensor integration, embedded firmware, and wireless notification capabilities into a manufacturable hardware design.

The system was developed not only as a functional weather monitoring device, but also as a full engineering design exercise involving schematic capture, PCB layout, debugging, manufacturing preparation, and system-level decision making.

Purpose

The purpose of this project is to create a smaller and more personalized alternative to large commercial weather stations while still providing meaningful environmental monitoring data to the user.

The system is designed to:

- Monitor environmental conditions in real time
- Provide simple automatic operation without user interaction
- Demonstrate embedded hardware and PCB design skills
- Explore future predictive weather analysis using regression models
- Prepare the design for future manufacturability and expansion

Possible future implementations include mobile app integration, wireless alerts, cloud logging, and weather prediction systems.

Objectives

The major objectives of the project were:

- Design a custom 4-layer PCB using KiCad
- Interface a BME280 environmental sensor with an ATmega328P-A microcontroller
- Measure temperature, humidity, and atmospheric pressure
- Create a manufacturable PCB design complete with Gerber and drill files
- Validate PCB functionality through proof-of-concept testing
- Explore future weather prediction using regression-based analysis
- Gain experience with real-world engineering design workflows

System Overview

The system uses an ATmega328P-A microcontroller as the primary processing unit and a BME280 sensor for environmental sensing through SPI communication.

The device can be powered either through:

- Screw terminal external power input
- Two AA batteries for portable operation

The current proof-of-concept implementation uses a Raspberry Pi and GUI for demonstration and visualization purposes. Sensor data is continuously read and displayed automatically once powered on.

Future revisions may replace the GUI with:

- Mobile application integration
- Wireless phone notifications using ntfy
- Cloud-connected monitoring systems

User Interaction Stories

Basic Operation

1. User powers on the device
2. Sensor automatically begins collecting environmental data
3. Data is processed and displayed to the user
4. Optional wireless notifications may alert the user of changing environmental conditions

Future Operation

Future versions may allow:

- App-based monitoring
- Remote environmental alerts
- Cloud logging
- Weather prediction notifications

User Interface

Current Proof-of-Concept:

- GUI displaying environmental sensor readings
- Real-time monitoring updates

Planned/Future Interfaces:

- Mobile app integration
- ntfy push notifications
- Remote monitoring interface

The final standalone PCB design itself does not require onboard buttons or displays for operation.

User Acceptance Criteria

Given:

The system is powered on with a functioning sensor connected

When:

The device begins operation

Then:

The system shall:

- Read environmental sensor data
- Process temperature, humidity, and pressure values
- Display environmental conditions
- Operate continuously without user interaction

Technical Parameters

Hardware

- ATmega328P-A microcontroller
- BME280 environmental sensor
- Custom 4-layer PCB
- SPI communication interface
- Screw terminal power input
- Battery power support (2 AA batteries)

Power

- External power through screw terminal
- Portable battery operation

PCB Design

- 4-layer stackup
- Manufacturable design
- ERC and DRC verified
- Designed in KiCad

Functions

Core system functions include:

- Environmental sensing
- Temperature monitoring
- Humidity monitoring
- Atmospheric pressure monitoring
- Real-time data processing
- Wireless notification capability
- Future predictive weather analysis

Integration

The system integrates:

- Environmental sensing hardware
- Embedded processing
- PCB manufacturing workflow
- GUI visualization
- Future wireless communication systems

Communication between the sensor and MCU is handled using SPI.

Operational Requirements

The system is designed for:

- environmental monitoring
- Continuous operation
- Low-power embedded sensing
- Portable or stationary use

Manufacturing & Life Cycle

The PCB design is fully prepared for manufacturing and assembly. Gerber files, drill files, BOM documentation, and PCB layout files are included in the repository for future fabrication.

The board has not yet been mass manufactured, but the design is considered manufacturing-ready and can be submitted directly to PCB fabrication services such as JLCPCB or PCBWay.

Future revisions may include:

- Custom enclosure development
- Waterproofing
- Improved sensor packages
- Commercial-grade deployment

Design Decisions

Several major design decisions were made during development:

- Transitioned from DHT22 to BME280 for improved sensing capability
- Moved to a 4-layer PCB after routing constraints and additional GPIO considerations created layout conflicts on a 2-layer design
- Added additional design review improvements and missing hardware components during PCB review
- Chose SPI communication for sensor integration
- Used a GUI-based Raspberry Pi proof-of-concept for live demonstrations while designing the final standalone embedded PCB separately

Individual Contributions – Joshua Chary

Primary contributions included:

- Project documentation
- Development logging
- Design review participation
- PCB review assistance
- Identifying and documenting missing design elements
- Organizing project decisions and engineering workflow documentation
- Assisting with overall project planning and presentation preparation

A major focus of contribution was documenting the engineering process, design revisions, debugging stages, and final project preparation.

Future Improvements

Potential future improvements include:

- Mobile app integration
- Advanced weather prediction using regression models
- Cloud data logging
- Waterproof enclosure design
- Higher-end environmental sensors
- Remote monitoring capability
- Expanded wireless communication features
- Mass-production enclosure development

Key Concerns

The most important design concerns included:

- Reliable environmental sensing

- Stable sensor communication
- Manufacturable PCB routing
- Expandability for future features
- Balancing affordability with functionality
- Power efficiency for portable operation
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